

Final Test Submission

Problem 1

a) Summarize the key points Hinton makes about neural networks and artificial intelligence. What are his main arguments or observations about the current state and trajectory of AI?

In the AI: What Could Go Wrong? video with Geoffrey Hinton, he explains that modern neural networks learn by adjusting the strength of connections across many layers. Unlike older systems such as search engines or traditional rule-based programming, neural networks do not rely on hand-written rules. Instead, they automatically discover useful features (such as edges, shapes, or even parts of objects) through these connection adjustments. Backpropagation makes it possible to update millions of parameters at once, but the real breakthrough came when researchers gained access to much greater computing power and massive datasets. With those resources, neural networks were finally able to uncover complex patterns at scale.

Hinton extends these ideas to large language models, describing how tokens are represented as distributed features and how interactions across layers allow the system to predict the next word. Human feedback after training further shapes their behavior. He emphasizes that these models resemble human cognition more than traditional rule-based software. He also notes that digital neural networks have a unique advantage: identical copies can be trained on different data and then share their updates. His concerns focus on both near-term misuse—such as targeted persuasion, election interference, or dangerous scientific guidance—and longer-term risks if systems surpass human

intelligence and use persuasion to resist control. He observes that Europe and China appear more proactive about existential risk and regulation, while the U.S. risks weakening its leadership by cutting back on basic research funding. At the same time, he predicts major workforce changes, with routine intellectual tasks likely to be automated, but also highlights the potential benefits of AI in areas such as healthcare, education, and materials discovery.

b) Reflect on your own perspective regarding the points raised in the video. Where do you agree or disagree with Hinton's views? Support your position with reasoned arguments.

I agree on the need for governance and guardrails so that AI is not misused in the wrong hands. Hinton's optimism about Europe and China leading existential-risk coordination seems partly pragmatic, since both have shown stronger regulatory impulses. In the long run, a lasting solution will only work if countries and companies cooperate, creating systems that make it in everyone's interest (both governments and industry) to follow the same rules and safeguards. I also strongly agree that sustained basic research funding is one of the most reliable ways to ensure national resilience and advance safety science. Reducing public research makes the system weaker and lets private companies set the agenda based on their own limited goals.

I find Hinton's technical points convincing: representation learning through deep architectures, emergent feature hierarchies, and scaling laws in data and compute are the foundation of recent progress. His view that large language models are "more like us" than rule-based systems makes sense, since they rely on distributed representations and learned inference rather than fixed rules. I also share his caution about persuasion risk. These models already personalize messaging, and when combined with behavioral data

they become powerful tools for influence. Where I am more cautious is in the leap from “better sharing and persuasion” to near-term scenarios of AI fully “taking over.” Alignment and control are difficult problems, but real-world agency still depends on access to resources, infrastructure, and coordinated execution. The main obstacles are still tied to institutions and social systems. Even so, Hinton’s warning that smarter AI could use persuasion to avoid control seems realistic enough that strong safeguards are needed.

I do not think progress should be stopped, but I do worry about this technology falling into the wrong hands. I also don’t believe that prospering tech companies alone should be in control. There should be some form of government system that provides checks and balances, ensuring that AI development benefits society while minimizing the risk of harm.

c) Based on the video and your understanding of the field, what developments do you anticipate in artificial intelligence over the next 2-5 years? Consider technological advances, societal implications, or challenges that may emerge. Your predictions should be grounded in current trends and your analysis of the field. Technological advances

Based on Hinton’s discussion and current trends in the field, I expect AI to continue advancing quickly over the next few years. As Hinton mentioned in the video, AI is “immortal.” The hardware can be taken apart, but as long as the data is saved, it can be placed into new hardware and still remember everything. As far as the next 2-5 years, I believe these changes will happen both on the technical side and in how society adapts to the technology.

In terms of technological advances, multimodal models will allow AI systems to work with text, code, images, audio, video, and even actions all together. By combining these different inputs with tools and memory, they will be better at planning, finding accurate information, and avoiding errors in important situations. Smaller, efficient models

created through methods like distillation and retrieval-augmented generation will make AI systems lighter and faster. This means strong assistants could run directly on personal devices, giving private, quick support and specialized tools. It's similar to what Hinton mentioned about our phones predicting text. AI will also improve at handling multi-step tasks such as research, data analysis, or coding. These assistants that can carry out tasks will be made safer by using controlled environments, rules, and checks to make sure everything works correctly.

Safety, governance, and industry shifts will also play a major role. Companies will start using standard tests to check for risks like persuasion, misuse in science, or too much autonomy. Safety checks will continue even after models are released, with updates added over time. Regulation will expand, but not evenly across countries. The EU is expected to push forward risk-based frameworks, while the U.S. may focus more on election security, watermarking content, and protecting infrastructure. International cooperation will likely focus on preventing catastrophic risks rather than sharing advanced capabilities. Training will also rely more on licensed datasets and synthetic data pipelines. This shift will lower legal risks and build more trust in how models are developed.

Societal challenges will emerge alongside these advances. I believe AI will quickly impact repetitive intellectual work such as writing reports, customer support, and documentation. Simultaneously, new jobs will grow in areas like oversight, domain expertise, and AI-assisted roles. In educational establishments, teachers will adapt by emphasizing process, critique, and originality. Personalized messaging will challenge information ecosystems, and systems for verifying content (such as signatures or credentials) will become essential, especially during election cycles. Efficiency improvements will help with compute and energy demands, but training large models will still require significant energy. Companies will need to invest in materials that will not impact the environment as negatively to mitigate further damage.

In conclusion, I expect a dual track. On one side, AI capabilities will keep accelerating and spreading into everyday workflows. On the other, safety, governance, and provenance systems will race to catch up. Whether AI stays positive overall will depend on steady support for public research, clear testing, and strong safeguards against misuse.

Problem 2

a) Explain the key concepts underlying the Transformer architecture. What fundamental mechanisms and design principles make this architecture distinctive? Include a discussion of the attention mechanism and how it differs from previous approaches.

The Transformer, introduced by Vaswani et al. (2017), represented a major departure from recurrence and convolution in sequence modeling. Its main innovation is the use of self-attention as the only mechanism for connecting elements in a sequence. Unlike recurrent neural networks (RNNs), which process tokens one by one and struggle with long dependencies, self-attention lets every token attend to every other token in a single step. Thus, reducing the path length between distant tokens to a constant, making it much easier to capture long-range relationships.

The scaled dot-product attention computes similarity scores between queries and keys, scales them by $\sqrt{dk}$ to avoid vanishing gradients, and applies a softmax to produce weights over values. This then allows the model to focus on the most relevant parts of the input. Multi-head attention extends this idea by projecting queries, keys, and values into multiple subspaces, so the model can capture different types of relationships at the same time. Due to the model having no recurrence or convolution, the model has to be told the position of each token. Vaswani et al. used sinusoidal positional encodings, which help the model generalize to sequence lengths longer than those seen during training.

Each encoder and decoder block also includes a feed-forward network with two linear layers separated by a ReLU activation. Training stability comes from residual connections and layer normalization, while dropout ($p = 0.1$) and label smoothing ($\varepsilon = 0.1$) provide regularization. Optimization used the Adam optimizer ($\beta_1 = 0.9$, $\beta_2 = 0.98$, $\varepsilon = 10^{-9}$) with a learning rate schedule that warmed up for 4000 steps before decaying with the inverse square root of the step number. Using these techniques, the Transformer achieved 28.4 BLEU on English-German and 41.0 BLEU on English-French, trained in just 3.5 days on 8 GPUs.

b) Describe the encoder and decoder components of the Transformer architecture. How do these components function, and what role does each play in sequence-to-sequence tasks? What are the key differences between the encoder and decoder blocks?

The Transformer follows the encoder-decoder design common in sequence-to-sequence tasks like machine translation.

The encoder is made up of six layers. Each layer has two parts: multi-head self-attention and a feed-forward network (linear $\rightarrow$ ReLU $\rightarrow$ linear). To keep training stable, residual connections and layer normalization are added around each part, and dropout is used on embeddings and outputs. The encoder's job is to turn the input sequence into richer representations, where each token is updated with information from all the others. Training used the Adam optimizer with a 4000-step warmup, which helped the model learn smoothly.

The decoder also has six layers, but each one has three parts: masked multi-head self-attention, encoder-decoder attention, and a feed-forward network (linear $\rightarrow$ ReLU $\rightarrow$ linear). The masking makes sure that predictions for a token only depend on earlier tokens, which preserves the autoregressive property of language generation. The encoder-decoder attention lets the decoder use the encoder's output, aligning the input and output sequences. As with the encoder, residual connections and layer normalization are applied

around each sub-layer, and dropout is used to regularize training. The feed-forward networks again use ReLU activations. Training relied on the Adam optimizer ($\beta_1 = 0.9$, $\beta_2 = 0.98$, $\epsilon = 10^{-9}$) with a 4000-step warmup schedule, which helped stabilize learning and improve convergence.

The key differences between the encoder and decoder are that the encoder relies solely on self-attention to build input representations, while the decoder combines self-attention with cross-attention to the encoder outputs. The decoder also applies masking to prevent information leakage from future tokens, ensuring autoregressive generation. In this way, the encoder produces a detailed representation of the input, and the decoder uses it to generate the output step by step.

c) Analyze the impact of the Transformer architecture on the field of machine learning and natural language processing. Why has this architecture been so influential? What capabilities or advantages does it provide that previous architectures lacked?

The Transformer has been a huge influence in machine learning and natural language processing. By removing recurrence, it enabled massive parallelization during training, which cut training time compared to RNNs while improving accuracy. The base model trained in just 12 hours on 8 GPUs already surpassed prior state-of-the-art systems. Techniques such as ReLU activations, residual connections, and layer normalization stabilized optimization, while regularization methods like dropout ($p = 0.1$) and label smoothing ($\epsilon = 0.1$) further improved performance. Combined with the Adam optimizer ($\beta_1 = 0.9$, $\beta_2 = 0.98$, $\epsilon = 10^{-9}$) and a 4000-step warmup schedule, these methods made training both robust and efficient.

The Transformer also shows that it is highly scalable. Later models built on the Transformer, such as BERT (Devlin et al., 2018), GPT (Radford et al., 2018), and T5 (Raffel et al., 2019), showed that scaling model size and training data yields dramatic gains,

encouraging the rise of large language models. The authors highlighted that attention models could be used beyond text, in areas like images, audio, and video. They suggested using local attention to handle very large inputs more efficiently. This means that the Transformer was a flexible design that could work across many types of data, and it later became a foundation for AI.

The architecture also offered new interpretability, since attention weights provide insight into what the model focuses on, offering more transparency than recurrent networks. Most importantly, the Transformer shifted research priorities: instead of designing new architectures for each task, researchers now focus on scaling, pretraining, and fine-tuning. This shift made powerful models easier to use and speed up progress in NLP, from translation to chatbots to code generation.

In summary from these notable observations, the Transformer removed the limits of recurrence, made scaling possible, and became the foundation of today's generative AI, changing how models are built and applied across different domains.